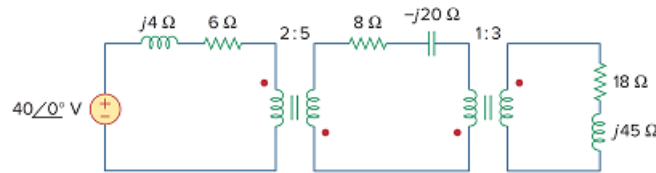


Problem

For the network in Fig. 13.127, find:

- (a) the complex power supplied by the source,
 (b) the average power delivered to the 18- Ω resistor.

Figure 13.127



Step-by-step solution

Step 1 of 9

(a)

Refer to Figure 13.127 in the textbook. Calculate the value of Z_L .

$$Z_L = (18 + j45) \Omega$$

Calculate the value of the reflected impedance in the middle circuit from the load.

$$Z_L' = \frac{Z_L}{n^2} \dots\dots (1)$$

Calculate the value of n .

$$n = \frac{N_2}{N_1}$$

Substitute 3 for N_2 and 1 for N_1 .

$$\begin{aligned} n &= \frac{3}{1} \\ &= 3 \end{aligned}$$

Step 2 of 9

Substitute $(18 + j45) \Omega$ for Z_L and 3 for n in the equation (1).

$$\begin{aligned} Z_L' &= \frac{(18 + j45)}{(3)^2} \\ &= \frac{(18 + j45)}{9} \\ &= (2 + j5) \Omega \end{aligned}$$

The total impedance in the middle circuit is,

$$\begin{aligned} Z_m &= (8 - j20) + (2 + j5) \\ &= (10 - j15) \Omega \end{aligned}$$

Step 3 of 9

Calculate the value of the reflected impedance in the primary circuit from the middle circuit.

$$Z_p' = \frac{Z_m}{n^2} \dots\dots (2)$$

Calculate the value of n .

$$n = \frac{N_2}{N_1}$$

Substitute 5 for N_2 and 2 for N_1 .

$$\begin{aligned} n &= \frac{5}{2} \\ &= 2.5 \end{aligned}$$

Step 4 of 9

Substitute $(10 - j15) \Omega$ for Z_m and 2.5 for n in the equation (2).

$$\begin{aligned} Z_p' &= \frac{(10 - j15)}{(2.5)^2} \\ &= \frac{(10 - j15)}{(6.25)} \\ &= (1.6 - j2.4) \Omega \end{aligned}$$

The total impedance in the primary circuit is,

$$\begin{aligned} Z_{in} &= (6 + j4) + (1.6 - j2.4) \\ &= (7.6 + j1.6) \Omega \\ &= 7.77 \angle 11.89^\circ \Omega \end{aligned}$$

Step 5 of 9

Calculate the current in the primary circuit.

$$\begin{aligned} I_1 &= \frac{40 \angle 0^\circ}{Z_{in}} \\ &= \frac{40 \angle 0^\circ}{7.77 \angle 11.89^\circ} \\ &= 5.15 \angle -11.89^\circ \text{ A} \end{aligned}$$

Calculate the complex power supplied by the source.

$$S = V_s I_1^*$$

Substitute $40 \angle 0^\circ \text{ V}$ for V_s and $5.15 \angle 11.89^\circ \text{ A}$ for I_1^* .

$$\begin{aligned} S &= (40 \angle 0^\circ)(5.15 \angle 11.89^\circ) \\ &= 206 \angle 11.89^\circ \text{ VA} \end{aligned}$$

Hence, the complex power supplied by the source is $206 \angle 11.89^\circ \text{ VA}$.

Step 6 of 9

(b)

Calculate the current in the second loop.

$$\begin{aligned} I_2 &= \frac{-I_1}{n} \\ &= \frac{-I_1}{\left(\frac{N_2}{N_1}\right)} \end{aligned}$$

Step 7 of 9

Substitute $5.15\angle-11.89^\circ$ A for I_1 , 5 for N_2 and 2 for N_1 .

$$\begin{aligned} I_2 &= \frac{-(5.15\angle-11.89^\circ)}{\left(\frac{5}{2}\right)} \\ &= -2.06\angle-11.89^\circ \text{ A} \end{aligned}$$

Step 8 of 9

Calculate the current in the third loop.

$$\begin{aligned} I_3 &= \frac{-I_2}{n} \\ &= \frac{-I_2}{\left(\frac{N_2}{N_1}\right)} \end{aligned}$$

Substitute $-2.06\angle-11.89^\circ$ A for I_2 , 3 for N_2 and 1 for N_1 .

$$\begin{aligned} I_3 &= \frac{-(-2.06\angle-11.89^\circ)}{\left(\frac{3}{2}\right)} \\ &= -0.69\angle-11.89^\circ \text{ A} \end{aligned}$$

Step 9 of 9

Calculate the average power delivered to the $18\ \Omega$ resistor.

$$P_{18\ \Omega(\text{avg})} = |I_3|^2 R$$

Substitute $-0.69\angle-11.89^\circ$ A for I_3 and $18\ \Omega$ for the R .

$$\begin{aligned} P_{18\ \Omega(\text{avg})} &= |-0.69|^2 (18) \\ &= 8.57 \text{ W} \end{aligned}$$

Hence, the average power across the $18\ \Omega$ resistor is 8.57 W .